

Lab 1 – Characterisation of the protocol efficiency

2025/2026

Lab Report Structure

Objective

The purpose of this report is to present an **efficiency analysis** of the data link protocol implemented during the first laboratory work of the *Computer Networks* course.

Introduction

The main objective of the lab is to design and implement a reliable communication protocol that operates over a serial RS-232 link between two Linux-based computers. Following the principles of the data link layer, the implemented protocol ensures reliable data transfer through framing, error detection, acknowledgements, and retransmissions using the Stop-and-Wait Automatic Repeat Request (ARQ) mechanism. The system integrates a transmitter and a receiver application that exchange data and control packets over this link, supporting file transfer operations while maintaining layer independence.

This report should focus on **evaluating the efficiency (S)** of the implemented protocol under different transmission conditions. Parameters such as the **Frame Error Rate (FER)**, **propagation delay (T_{prop})**, **Baudrate (C)** and **frame size (L)** can be changed, allowing the comparison between the obtained results and theoretical efficiency models.

Suggestions

- **Study the theoretical formulas** for efficiency of ARQ protocols (check the 03-data-link-layer.pdf slides from theoretical classes);
- **Measure the file transference time** varying the FER, T_{prop} , C and L. Vary one parameter at a time;
- To vary FER, you can randomly generate errors on I frames;
- To vary T_{prop} a propagation delay can be considered.
- For each experience, calculate the **achieved throughput R**, and calculate the **measured efficiency S**;
- **Plot the efficiency S for each set of parameters (FER, T_{prop} , C and L) and discuss the results**;
- Compare the experimental results with the known theoretical formulas for efficiency, and check the validity of the formulas.

Note: you can use `cable.c` to perform the protocol efficiency evaluation. The program contains options to change the baudrate, the bit error rate (BER) and the propagation delay. Read the program instructions that are printed in the terminal when the program starts.

Structure

The report should contain the following information:

- Title, Authors
- Summary
- Introduction
- Methodology
- Results
- Conclusions

Formatting Guidelines

- Length: **3 pages maximum**, including figures and tables
- File format: PDF, named “Authros_Lab1_Report.pdf”
- Language: Portuguese or English
- Font size 11pt
- Identify the author names and their class on the first page
- Figures and tables must be numbered and captioned
- Include references only if external sources are cited (optional)

Writing Tips

- Structure your text by paragraphs. Each paragraph should focus on one clear topic. The first sentence must make that topic explicit;
- Keep paragraphs compact and purposeful. Aim for a clear and visually organized layout that helps quick reading;
- Write positively. Describe what you implemented, measured, or observed, not what you did not do;
- Be precise. Avoid vague terms like fast or many. Use measurable facts instead (e.g., the function executes in 200 ms or the code has 800 lines);
- Be concise. After writing, reread and simplify. If you can say the same thing in fewer words, do it;
- Avoid redundancy. Do not restate the same idea in multiple ways; rewrite for clarity instead;
- Maintain logical flow. Introduce each concept before using it, and always use consistent terms for the same idea (e.g., frame, packet, efficiency).